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RAPTOR: Robust and Perception-Aware Trajectory Replanning for Quadrotor Fast Flight

Boyu Zhou, Jie Pan, Fei Gao, and Shaojie Shen

***Abstract*—Recent advances in trajectory replanning have enabled quadrotor to navigate autonomously in unknown environments. However, high-speed navigation still remains a significant challenge. Given very limited time, existing methods have no strong guarantee on the feasibility or quality of the solutions. Moreover, most methods do not consider environment perception, which is the key bottleneck to fast flight. In this article, we present RAPTOR, a robust and perception-aware replanning framework to support fast and safe flight, which addresses these issues systematically. A path-guided optimization approach that incorporates multiple topological paths is devised, to ensure finding feasible and high-quality trajectories in very limited time. We also introduce two perception-aware planning approaches to actively observe and avoid unknown obstacles. A risk-aware trajectory refinement ensures that unknown obstacles which may endanger the quadrotor can be observed earlier and avoid in time. The motion of yaw angle is planned to actively explore the surrounding space that is relevant for safe navigation. The proposed methods are tested extensively through benchmark comparisons and challenging indoor and outdoor aggressive flights. We release our implementation as an open-source package[1] for the community.**

***Index Terms*—Aerial systems, perception and autonomy, collision avoidance, motion and path planning, trajectory planning.**

I. Introduction

IN RECENT years,[1] progresses on different aspects of unmanned aerial vehicles (UAVs), especially quadrotor autonomy, have been achieved, which promote autonomous navigation. Nonetheless, high-speed flight in unknown and highly cluttered environments still remains one of the biggest challenges toward full autonomy. To achieve fast flight, trajectory replanning is of vital importance to cope with previously unknown obstacles, guaranteeing smooth and safe navigation.

Although trajectory replanning has been investigated actively, most presented methods only apply to flights at a moderate speed. Several issues greatly hinder their usage in high-speed scenarios. (a) Flying in unknown environments at a high speed, the quadrotor should replan new trajectories to avoid unexpected obstacles in considerably short time, otherwise it crashes. However, most methods do not guarantee to find feasible trajectories given very limited time. (b) Current method typically find a single trajectory confined within a topologically equivalent class, which can be easily trapped in local minima and lead to unsatisfactory flight smoothness and safety. (c) Existing methods are unaware of environment perception, which can be fatal as the flight speed and obstacle density get high. Paying no attention to perception, the planned motions may lead to restricted visibility to the environments, which would in turn result in deficient information of the surrounding space necessary for safe navigation. The consequence of not considering perception in replanning can be better illustrated by Fig. 1. To minimize the energy consumption, a trajectory near the wall is generated, along which the visibility toward the unknown space behind the corner is very limited. As a result, the obstacle right behind the corner is invisible until the quadrotor turns right and gets very close, which "surprises" or even crashes the quadrotor. Instead of avoiding what are observed passively, actively observing and avoiding possible dangers is critical for safe high-speed flight.

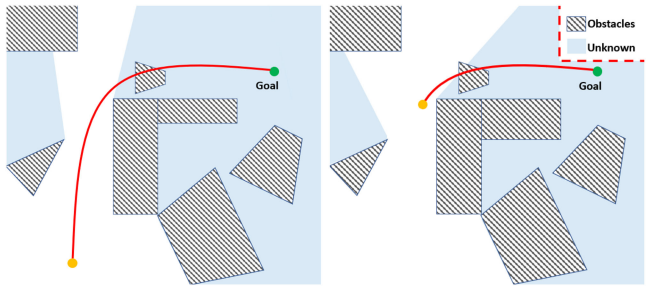


Fig. 1. An example of planning without perception awareness. The quadrotor flies near the wall, where it has poor visibility to the space behind the corner. In consequence, an obstacle is not revealed until the quadrotor gets very close.

In this article, we propose a **R**obust **A**nd **P**erception-aware **T**raject**O**ry **R**eplanning framework called **RAPTOR** to address these issues systematically. To ensure obtaining feasible trajectories within limited time, we present a path-guided gradient-based optimization method, which utilizes geometric guiding paths to eliminate infeasible local minima and guarantee the success of replanning. Also, to prevent getting stuck in local minima and further improve the quality of replanned trajectories, we introduce an online topological path planning to extract a comprehensive set of distinctive paths to guide the optimization.

Manuscript received July 4, 2020; accepted March 19, 2021. This work was supported in part by RGC General Research Fund under Project 16213717 and in part by HKUST Institutional Fund. *(Corresponding author: Boyu Zhou.)*

Boyu Zhou and Shaojie Shen are with the Department of Electronic and Computer Engineering, Hong Kong University of Science and Technology, Hong Kong (e-mail: 405540572@qq.com; eeshaojie@ust.hk).

Jie Pan is with the Hong Kong Center for Construction Robotics, Hong Kong University of Science and Technology, Hong Kong (e-mail: jpanai@connect.ust.hk).

Fei Gao is with the State Key Laboratory of Industrial Control and Technology, Zhejiang University, Hangzhou 310007, China (e-mail: fgaoaa@zju.edu.cn).

This article has supplementary material provided by the authors and color versions of one or more figures available at https://doi.org/10.1109/TRO.2021.3071527.

Digital Object Identifier 10.1109/TRO.2021.3071527

[1]Code is available at github.com/HKUST-Aerial-Robotics/Fast-Planner

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With the guidance, multiple trajectories that pass through diverse collision-free regions are explored, which leads to a more thorough search of the solution space and provides superior replanning compared with generating a single trajectory.

The above method addresses issues (a) and (b). However, it adopts the optimistic assumption and lacks the awareness of environment perception. To bridge this gap, two perception-aware planning approaches are proposed to enable faster and safer flight. First, a risk-aware trajectory refinement approach is developed to incorporate with the optimistic planner. It identifies unknown regions along the optimistic trajectories that are potentially dangerous to the quadrotor. Visibility toward such regions along with safe reaction distance are enforced explicitly, ensuring that obstacles in unmapped areas become visible earlier and are avoidable by the quadrotor. Second, we incorporate the yaw angle of the quadrotor into a two-step motion planning framework. An optimal sequence of yaw angles that maximizes information gain and smoothness is searched in the discrete state-space, which is further smoothed through optimization. The planned motions of yaw angle enable the quadrotor whose field-of-view (FOV) is limited to actively explore the unknown space to gain more relevant knowledge for the future flight.

Our method is systematically evaluated in both simulated and real-world experiments. Comprehensive benchmark comparisons against state-of-the-art methods show the superior performance of our method in several aspects in fast flight tasks. In particular, the proposed method is able to support fast and safe flight in challenging scenarios with limited visibility, while previous methods may fail to ensure safety. Moreover, extensive aggressive flight tests in complex indoor and outdoor environments are conducted to validate our planning system.

The topological path-guided optimization (PGO) is first proposed in our prior work [1] (Sections IV and V). Although it generates high-quality trajectories in short computation time, it is ignorant to environment perception, restricting its capability in higher speed and more cluttered environments. Particularly, some "surprising" obstacles crash the quadrotor occasionally, due to poor visibility and inadequate reaction time. In this article, we extend it by two perception-aware planning approaches (Sections VI–VII), enabling the quadrotor to perceive and avoid "surprising" obstacles actively. Thanks to it, dangers can be discovered substantially earlier, giving the quadrotor more confidence to navigate in high speed. Besides, more experiments of aggressive flight are conducted in a wider variety of challenging environments, to validate the performance of our overall replanning framework comprehensively. The contributions of this article are summarized as follows.

1) A risk-aware trajectory refinement, which enforces visibility and safe reaction distance to unknown obstacles. It improves the predictability and safety of fast flights.

2) A two-step yaw angle planning method, to actively explore the unknown environments and gather useful information for the flight.

3) Integration of (1) and (2) with our prior topological PGO approach [1], forming a complete replanning framework. Extensive simulation and real-world tests that validate each component and the overall framework. The source code of our system will be released.

In what follows, we review related works in Section II and introduce our system in Section III. Our path-guided trajectory optimization and topological path planning are detailed in Sections IV and V. The risk-aware trajectory refinement and yaw angle planning are presented in Sections VI and VII, respectively. Benchmark and experimental results are given in Section VIII. Section IX concludes this article.

II. Related Work

A. Quadrotor Trajectory Planning

Trajectory planning for quadrotor has been widely investigated. Existing methods can be categorized into hard-constrained methods and gradient-based optimization methods. Hard-constrained methods are pioneered by minimum-snap trajectory [2], in which piecewise polynomial trajectories are generated through quadratic programming (QP). Richter *et al.* [3] presented a closed-form solution to minimum snap trajectories. Refs. [4]–[8] generate trajectories in a two-step pipeline. Safe regions around initial paths are extracted as convex flight corridors, within which QP is solved to generate smooth and safe trajectories. Among these methods, poorly chosen time allocation of piecewise polynomials usually lead to unsatisfying results. To this end, fast marching [5] and kinodynamic search [8], [9] are utilized to search for initial paths with more reasonable time allocation. Gao *et al.* [5] also proposed to represent trajectories as piecewise Bézier curves so that safety and dynamical feasibility are guaranteed. Tordesillas *et al.* [10] adopted a mixed-integer QP (MIQP) formulation to find a more reasonable time allocation of the trajectory.

Another category is the gradient-based trajectory optimization (GTO) methods, which typically formulate trajectory generation as nonlinear optimization problems and incorporate the Euclidean signed distance field (ESDF) to ensure safety. Recent works [1], [11]–[15] revealed that they are particularly effective for local replanning, which is a key component for high-speed flight in unknown environments. Ratliff *et al.* [16] proposed to optimize discrete-time trajectory through covariant gradient descent, reviving the community's interest in such methods. Kalakrishnan *et al.* [17] presented a similar formulation, but solves the problem by sampling neighboring candidates iteratively. The stochastic sampling strategy partially overcomes the local minima issue but is computationally intensive. Oleynikova *et al.* [11] extended it to continuous-time quadrotor trajectories and also adopted random optimization restarts to slightly relieve the typical local minima issue of such methods. In [12], [13], the optimization is combined with an informed sampling-based path searching to improve the success rate. Usenko *et al.* [14] proposed to parameterize trajectories as uniform B-splines, showing the usefulness of continuity and locality properties of B-spline for replanning. However, due to insufficient success rate and efficiency, [11], [12], [14] only apply to flight at a moderate speed. To this end, Zhou *et al.* [15] further exploited B-spline to improve the efficiency and robustness. GTO methods are preferable for replanning due to high efficiency. However, their

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local minima issue may lead to undesired solutions. Our method lies in this category, and we resolve local minima by introducing topologically distinctive paths to guide optimization in parallel.

B. Topological Path Planning

There are works utilizing the idea of topologically distinct paths for planning, in which paths belonging to different homotopy (homology) [18]–[22] classes or visibility deformable groups [23] are sought. Schmitzberger *et al.* [18] construct a variant of probabilistic roadmap (PRM) to capture homotopy classes, in which path searching and redundant path filtering are conducted simultaneously. In contrast, [19], [20] first creates a PRM or Voronoi diagram, after which a homology equivalence relation based on complex analysis [21] is adopted to filter out redundant paths. These methods only apply to 2D scenarios. To seek for 3D homology classes, Bhattacharya *et al.* [22] exploit the theory of electromagnetism and propose a 3D homology equivalence relation. However, it requires occupied space to be decomposed into "genus 1" obstacles, which is usually impractical. Besides, capturing only homotopy classes in 3D space is insufficient to encode the set of useful paths, as indicated in [23], since 3D homotopic paths may be too hard to deform into each other. To this end, Jaillet and T. Siméon [23] leverage a visibility deformation roadmap to search for a richer set of useful paths. Oleynikova *et al.* [24] and Blochliger *et al.* [25] convert maps built from SLAM systems into sparse graphs representing the topological structure of the environments. However, they rely on processing massive voxels of the entire map to build the sparse graphs, which is too expensive for online usage. Our topological path searching is conceptually closest to [23], but we extend the concept of visibility deformation to make it computationally cheaper, and introduce a path shortening and pruning technique to maintain compact sets of useful paths.

C. Navigation in Unknown Environments

To deal with unknown environments in navigation, different strategies have been used. Many methods adopt the optimistic assumption [7], [14], [15], [26], which treats the unknown space as collision-free. This strategy improves the chance of reaching goals, but may not guarantee safety. On the contrary, some other methods regard unknown space as unsafe and only allow motions within the known-free space [27] or sensor FOV [28], [29]. In [30], the sensor FOV constraint is partially relaxed by choosing safe motion primitives generated in the past. Although these restrictions ensure safety, they lead to conservative motion. Recently, Tordesillas *et al.* [10] proposed a strategy that plans in both the known-free and unknown space. Instead of being over optimistic about the unknown space, it always maintains back-up trajectories to ensure safety.

The limitation of the above mentioned strategies is the lack of environment perception awareness, which is of significant necessity in fast flight. Although much attention has been paid to planning with the awareness of localization [31]–[33] and target tracking [34], [35], less emphasis is put on environment perception. Richter and Roy [36] proposed a learned heuristic function to guide the path searching into areas with greater visibility to unknown space, but it may not generalize well to complex 3D environments. Heiden *et al.* [37] showed an integrated mapping and planning framework for active perception. The planner iteratively simulates future measurements after executing specific motions, predicts uncertainty of the map, and minimizes the replanning risk. Its main drawback is the prohibited runtime for online usage. In [38], a local planner is coupled with local exploration to safely navigate a cluttered environment. However, it conservatively selects intermediate goals within known-free space, which restricts the flight speed. In this article, we present a perception-aware strategy to ensure that unknown dangers can be discovered and avoided early. It guarantees safety and does not lead to conservative behaviors.



Fig. 2. Composite image of a fast flight experiment in a cluttered unknown environment. Video is available at: https://youtu.be/6wrh4G1-cQ4.

III. System Overview

The proposed replanning system is illustrated in Fig. 3. It takes the outputs of the global planning, dense mapping, and state estimation modules, and deforms the global reference trajectory locally to avoid previously unknown obstacles. The replanning works in two steps. First, the robust optimistic replanning generates multiple locally optimal trajectories in parallel through the PGO (Section IV). The optimization is guided by topologically distinctive paths extracted and carefully selected from the topological path searching, which will be detailed in Section V. Optimistic assumption is adopted in this step. Second, the perception-aware planning strategy is utilized. The best trajectory among the locally optimal ones is further polished by a risk-aware trajectory refinement, in which its safety and visibility to the unknown and dangerous space are improved, as presented in Section VI. Based on the refined trajectory, the yaw angle is planned to actively explore the unknown environments (Section VII). The global planning, mapping, estimation, and controller are introduced briefly in Section VIII-A.

IV. Path-Guided Trajectory Optimization

As shown in Section II-A, GTO methods are effective for replanning, but suffer from local minima. To further improve the robustness of replanning and ensure flight safety, we present PGO, which utilizes a geometric guiding path in the optimization to guarantee its success.

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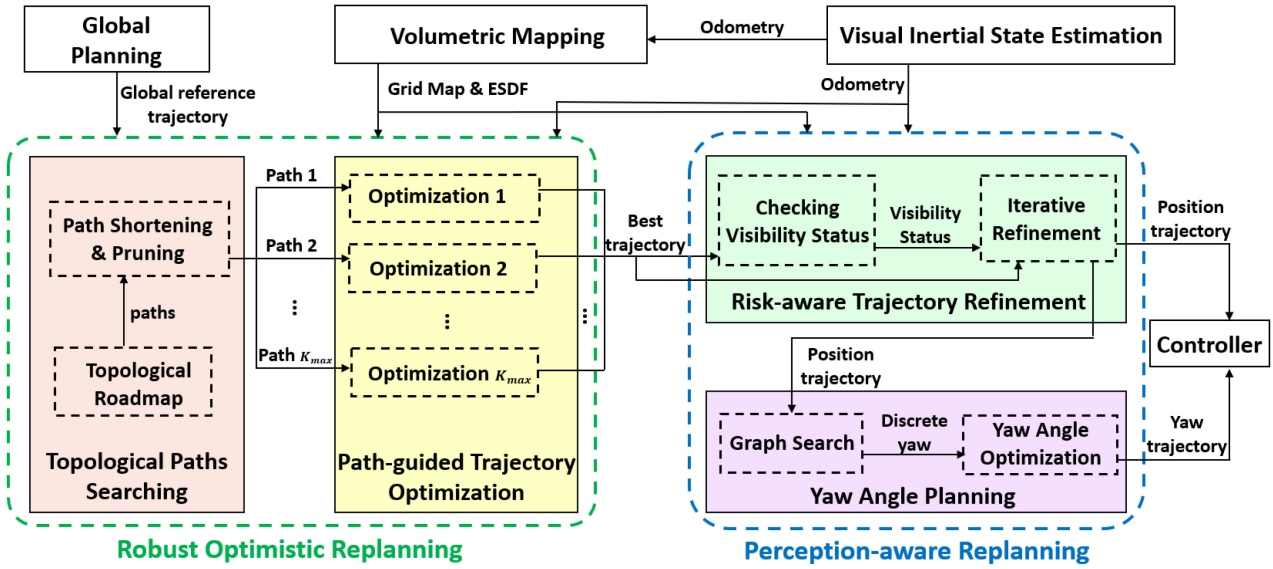


Fig. 3. An overview of our replanning system. It takes information from the global planning, mapping and state estimation and replans trajectory in two steps.

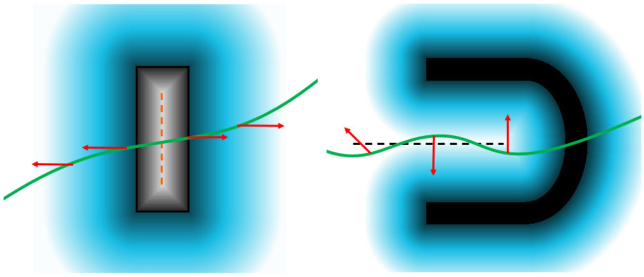


Fig. 4. Typical examples of optimization failure, where the trajectories pass through the "I"- and "U"-shaped obstacles. (a) Adjacent parts of the trajectory are pushed in opposing directions since it crosses the "valley" of the ESDF (denoted by dashed lines). Negative distance is in gray color, red arrows denote gradients of ESDF. (b) The trajectory crosses the "ridge".

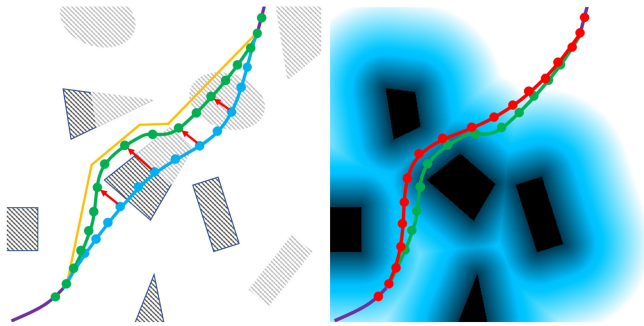


Fig. 5. Ttwo-phases PGO approach for trajectory replanning. (a) A geometric guiding path (yellow) attracts the original B-spline trajectory (blue) into the free space. obstacles are shown in transparent gray. (b) Warmup trajectory is further optimized in the ESDF to find a locally optimal trajectory (red).

A. *Optimization Failure Analysis*

Previous work [39] showed that failure of GTO is relevant to unfavorable initialization, i.e., initial paths that pass through obstacles in certain ways usually get stuck. Underlying reason for this phenomenon is illustrated in Fig. 4. Typical GTO methods incorporate the gradients of a ESDF in a collision cost to push the trajectory out of obstacles. Yet there are some "valleys" or "ridges" in the ESDF, around which the gradients differ greatly. Consequently, if a trajectory is in collision and crosses such regions, the gradients of ESDF will change abruptly at some points. This can make gradients of the collision cost push different parts of the trajectory in opposing directions and fail the optimization.

Normally, such "valleys" and "ridges," which correspond to the space that has an identical distance to the surfaces of nearby obstacles, are difficult to avoid, especially in complex environments. Therefore, optimization depending solely on the ESDF fails inevitably at times. To solve the problem, it is essential to introduce extra information that can produce an objective function whose gradients consistently deform the trajectory to the free space.

B. *Problem Formulation*

We propose PGO built upon our previous work [15] that represents trajectories as B-splines for more efficient cost evaluation. For a trajectory segment in collision, we reparameterize it as a p_b degree uniform B-spline with control points $\{\mathbf{q}_0, \mathbf{q}_1, \ldots, \mathbf{q}_N\}$ and knot span Δt.

PGO consists of two different phases. The first phase generates an intermediate *warmup trajectory*. As concluded above, external information should be included to effectually deform the trajectory, since solely applying the ESDF could be futile. We employ a geometric guiding path to attract the initial trajectory to the free space [depicted in Fig. 5(a)] since collision-free paths are readily available from standard methods like A* and RRT*. In our work, the paths are provided by the sampling-based topological path searching (Section V). The first-phase objective

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function is

$$\begin{aligned} f_{p1} &= \lambda_s f_s + \lambda_g f_g \\ &= \lambda_s \sum_{i=p_b-1}^{N-p_b+1} \left\| \mathbf{q}_{i+1} - 2\mathbf{q}_i + \mathbf{q}_{i-1} \right\|^2 \\ &\quad + \lambda_g \sum_{i=p_b}^{N-p_b} \left\| \mathbf{q}_i - \mathbf{g}_i \right\|^2 \end{aligned} \tag{1}$$

where f_s is the smoothness term, while f_g penalizes the distance between the guiding path and the B-spline trajectory. As in [15], f_s is designed as a elastic band cost function[2] that simulates the elastic forces of a sequence of springs. To simplify the design of f_g, we utilize the property that the shape of a B-spline is finely controlled by its control points. Each control point $\mathbf{q}_i$ is assigned with an associated point $\mathbf{g}_i$ on the guiding path, which is uniformly sampled along the guiding path. Then, f_g is defined as the sum of the squared Euclidean distance between these point pairs.

Notably, minimizing f_{p1} yields an unconstrained QP problem, so its optimal solution can be obtained in closed form. It outputs a smooth trajectory in the vicinity of the guiding path. Since the path is already collision-free, usually the warmup trajectory is also so. Even though it is not completely collision-free, its major part will be attracted to the free space. At this stage, the gradients of ESDF along the trajectory vary smoothly, and the gradients of the objective function push the trajectory to the free space in consistent directions. Hence, standard GTO methods can be utilized to improve the trajectory.

In the second phase, we adopt our previous B-spline optimization method [15] to further refine the warmup trajectory into a smooth, safe, and dynamically feasible one, whose objective function is

$$\begin{aligned} f_{p2} &= \lambda_s f_s + \lambda_c f_c + \lambda_d \left(f_v + f_a \right) \\ &= \lambda_s \sum_{i=p_b-1}^{N-p_b+1} \left\| \mathbf{q}_{i+1} - 2\mathbf{q}_i + \mathbf{q}_{i-1} \right\|^2 \\ &\quad + \lambda_c \sum_{i=p_b}^{N-p_b} \mathcal{F}(d(\mathbf{q}_i), d_{\min}) \\ &\quad + \lambda_d \sum_{\substack{\mu \in \\ \{x,y,z\}}} \left\{ \sum_{i=p_b-1}^{N-p_b} \mathcal{F}(v_{\max}^2, \dot{q}_{i,\mu}^2) + \sum_{i=p_b-2}^{N-p_b} \mathcal{F}(a_{\max}^2, \ddot{q}_{i,\mu}^2) \right\}. \end{aligned} \tag{2}$$

Here $\mathcal{F}()$ is a penalty function for general variables

$$\mathcal{F}(x,y) = \begin{cases} (x-y)^2 & x \le y \\ 0 & x > y \end{cases} \tag{3}$$

f_c is the collision cost that grows rapidly if the trajectory gets closer than $d_{\min}$ to obstacles, where $d(\mathbf{q})$ is the distance of point $\mathbf{q}$ in the ESDF. f_v and f_a penalize infeasible velocity and acceleration, where $\dot{\mathbf{q}}_i = [\dot{q}_{i,x}, \dot{q}_{i,y}, \dot{q}_{i,z}]^{\mathrm{T}}$ and $\ddot{\mathbf{q}}_i = [\ddot{q}_{i,x}, \ddot{q}_{i,y}, \ddot{q}_{i,z}]^{\mathrm{T}}$ are the control points of velocity and acceleration, which can be evaluated by

$$\dot{\mathbf{q}}_i = \frac{\mathbf{q}_{i+1} - \mathbf{q}_i}{\Delta t}, \quad \ddot{\mathbf{q}}_i = \frac{\mathbf{q}_{i+2} - 2\mathbf{q}_{i+1} + \mathbf{q}_i}{\Delta t^2} \tag{4}$$

$v_{\max}$ and $a_{\max}$ are single-axis maximum velocity and acceleration.[3] These two terms follow the formulations in [15], which utilizes the convex hull property of B-spline to guarantee dynamic feasibility. For any B-spline curve, each of its segments is completely confined within the convex hull of the associated control points. Moreover, derivatives of the curve are also B-splines, so the same property can be applied to bound the high-order derivatives. On these grounds, it suffices to constrain the control points of derivatives to ensure the dynamic feasibility of the entire trajectory.

Although PGO has one more step of optimization compared with previous methods, it can generate better trajectories within shorter time. On the one hand, the first phase is in unconstrained quadratic form, whose closed-form solution consumes negligible time. On the other hand, a warmup trajectory that is much easier to be refined is produced, which significantly shortens the convergence time compared with optimizing the initial trajectory from scratch.

V. Topological Path Searching

Given a single guiding path, our PGO method can obtain a locally optimal trajectory. However, during the optimization, the trajectory is deformed continuously and cannot pass across obstacles. As a result, it is restricted to a local area, and easily gets stuck in local minima. One solution to this limitation cloud be searching a kinodynamic path [15], [40], [41] directly near the global optimum. However, doing so takes nontrivial time, especially in complex environments.

Inspired by [20], we propose an efficient 3D topological path searching to facilitate the optimization. It finds a collection of distinctive paths, guiding parallel trajectory optimization in diverse topologies. Compared with a single trajectory, more ones distributed throughout the free space are searched, leading to superior solutions. Unlike previous methods [24], [25] which require processing a huge number of voxels, we adopt a computationally cheaper sampling-based approach as [18], [23]. Meanwhile, we redesign the topology relation to further speed up the computation.

A. Topology Equivalence Relation

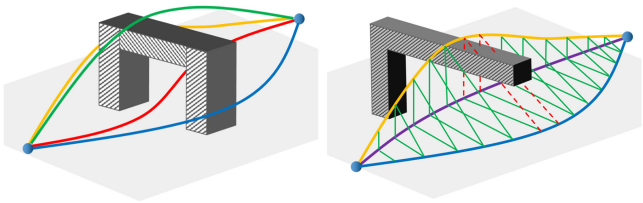
Although the concept of homotopy is widely used, it captures insufficient useful trajectories in 3D environments, as shown in Fig. 6(a). Jaillet and Siméon [23] propose a more useful relation in 3D space named visibility deformation (VD), but it is computationally expensive for deformation checking. Based

[2]Only the subset of control points $\{\mathbf{q}_{p_b}, \mathbf{q}_{p_b+1}, \ldots, \mathbf{q}_{N-p_b}\}$ is optimized due to the boundary state constraints of the trajectory. $\mathbf{q}_{p_b-2}, \mathbf{q}_{p_b-1}, \mathbf{q}_{N-p_b+1}$, and $\mathbf{q}_{N-p_b+2}$ are needed to evaluate the smoothness.

[3]It is desired that for each component of the velocity control point $\dot{\mathbf{q}}_i$, $\dot{q}_{i,\mu} \in [-v_{\max}, v_{\max}]$, or equivalently, $\dot{q}_{i,\mu}^2 \le v_{\max}^2$, where $\mu \in \{x,y,z\}$. It is similar for acceleration.

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(a) (b)

Fig. 6. (a) Green, blue, and yellow trajectories are equivalent by homotopy, but represent substantially different motions. (b) An illustration of UVD. The purple trajectory is deformable to the blue one by UVD, while the yellow trajectory is not deformable to the other two ones.

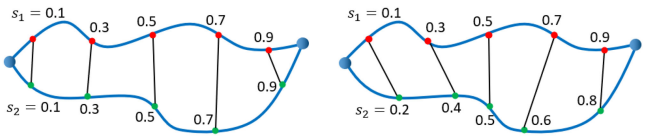


Fig. 7. Comparison between UVD (left) and VD (right). Each red point on one path is transformed to a green point on the other path. Any two associated points correspond to the same parameter s in UVD, but not in VD.

Algorithm 1: Topological Roadmap.

```
1  Initialize()
2  AddGuard(G, s), AddGuard(G, g)
3  while t ≤ t_max ∧ N_sample ≤ N_max do
4      p_s ← Sample()
5      g_vis ← VisibleGuards(G, p_s)
6      if g_vis.size() == 0 then
7          AddGuard(G, p_s)
8      if g_vis.size() == 2 then
9          path_1 ← Path(g_vis[0], p_s, g_vis[1])
10         distinct ← True
11         N_s ← SharedNeighbors(G, g_vis[0], g_vis[1])
12         for each n_s ∈ N_s do
13             path_2 ← Path(g_vis[0], n_s, g_vis[1])
14             if Deformable(path_1, path_2) then
15                 distinct ← False
16                 if Len(path_1) < Len(path_2) then
17                     Replace(G, p_s, n_s)
18                 break
19         if distinct then
20             AddConnector(G, p_s, g_vis[0], g_vis[1])
```

on VD, we define *uniform visibility deformation* (UVD), which also captures abundant useful trajectories, and is more efficient to check the existence of deformation.

Definition 1: Two trajectories $\tau_1(s)$, $\tau_2(s)$ parameterized by $s \in [0,1]$ and satisfying $\tau_1(0) = \tau_2(0), \tau_1(1) = \tau_2(1)$, are to each other by *visibility*, if for all s, line $\overline{\tau_1(s)\tau_2(s)}$ is collision-free.

Fig. 6(b) illustrates the relations of three trajectories based on UVD. The difference between VD and UVD is depicted in Fig. 7. Both of them define a continuous map between two paths $\tau_1(s)$ and $\tau_2(s)$, in which a point on $\tau_1(s)$ is transformed to a point on $\tau_2(s)$ through a straight line. The major difference is that for UVD, point $\tau_1(s_1)$ is mapped to $\tau_2(s_2)$, where $s_1 = s_2$, while for VD, s_1 does not necessarily equal s_2. Conceptually, both VD and UVD are specific kinds of homotopic transformation, capturing more useful paths in 3D space. UVD is less general than VD and characterizes particular subsets of VD. The definition of UVD provides a simpler way to check the existence of deformation,[4] making it more suitable for real-time replanning.

To test UVD, one can uniformly discretize $s \in [0,1]$ to $s_i = i/K, i = 0, 1, \ldots, K$ and check collision for lines $\overline{\tau_1(s_i)\tau_2(s_i)}$. For the piece-wise straight-line paths [as in Alg. 1, ()], we simply parameterize it uniformly, so that for any s except $\tau(s)$ is the join points of two straight lines, $\| \frac{d\tau(s)}{ds} \| = const.$

B. Topological Roadmap

To discover topologically distinctive paths, it is essential to create a skeleton graph [23]–[25] that captures the structure of the free space, in which multiple paths can be retrieved by standard graph search methods. To this end, Algorithm 1 is used to construct a UVD roadmap $\mathcal{G}$ capturing an abundant set of distinctive paths. Unlike standard PRM containing many redundant loops, our method generates a more compact roadmap where only a few paths may be deformable to others based on UVD [displayed in Fig. 8(a)–(c)].

We introduce two different kinds of graph nodes, namely *guard* and *connector*, similar to the visibility-PRM [42]. The guards are responsible for exploring different part of the free space, and any two guards g_1 and g_2 are not *visible* to each other (line $\overline{g_1 g_2}$ is in collision). Before the main loop, two guards are created at the start point s and end point g. Every time a sampled point is invisible to all other guards, a new guard is created at this point (Lines 6 and 7). To form paths of the roadmap, connectors are used to connect different guards (Lines 7–19). When a sampled point is visible to exactly two guards, a new connector is created, either to connect the guards to form a topologically distinct connection (Lines 19 and 20), or to replace an existing connector to make a shorter path (Lines 16 and 17). Limits of time ($t_{\max}$) or sampling number ($N_{\max}$) are set to terminate the loop.

With the UVD roadmap, a depth-first search augmented by a visited node list is applied to search for the paths between s and g, similar to [20].

C. Path Shortening and Pruning

As shown in Fig. 8(d), some paths obtained from Algorithm 1 may be detoured. Such paths are unfavorable, since in the first phase of PGO, it can deform a trajectory excessively and make it unsmooth. Hence, Algorithm 2 finds a shortcut path P_s that is deformable to each path P_r found by the depth-first search (illustrated in Fig. 9). The algorithm uniformly *discretizes* P_r to a set of points P_d. In each iteration, if a point p_d in P_d is invisible from the last point in P_s (Lines 3 and 4), the center of the first occupied voxel blocking the view of P_s.**back**() is found (Line 5). This point is then pushed away from obstacles

[4]To test the existence of VD, one should compute a visibility diagram and do path searching within it [23] which has higher complexity than testing UVD.

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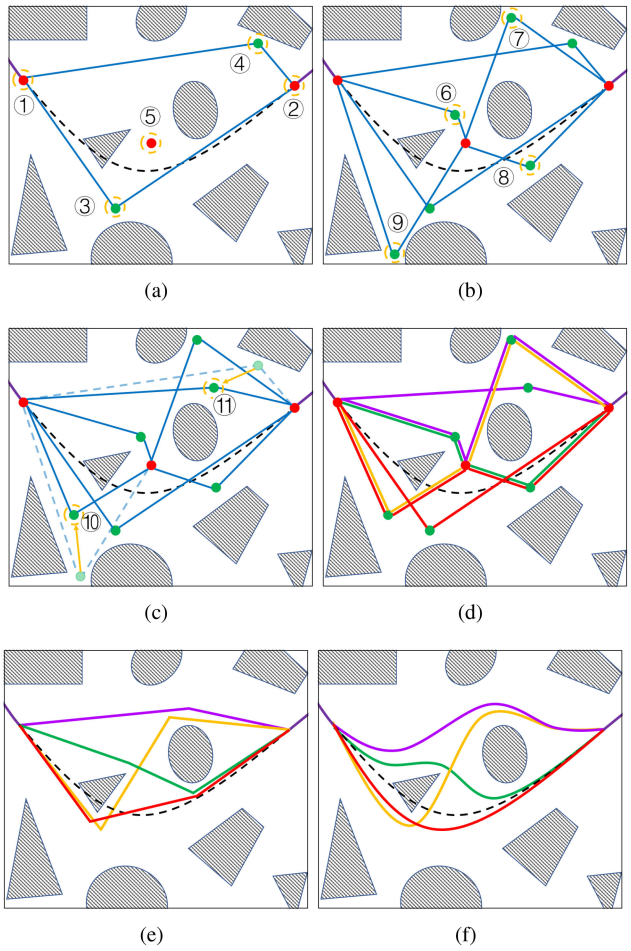


Fig. 8. Topological path searching and parallel trajectory optimization. (a) Two *connectors* (green) are sampled to connect the two initial *guards* (red) at the start and end points, after which a new *guard* is added to occupy a region not covered by others. (b) More *connectors* are created to connect the *guards*. (c) New *connectors* replace the old ones, making some connections shorter. (d) Paths are extracted from the roadmap. Paths that are deformable to each other are of the same color. (e) Extracted paths are shortened by Algorithm 2. (d) The shortened paths guide PGOs in parallel to generate several locally optimal trajectories. Note that the red trajectory is the smoothest since it takes less turns, while the associated guiding path is longer than others.

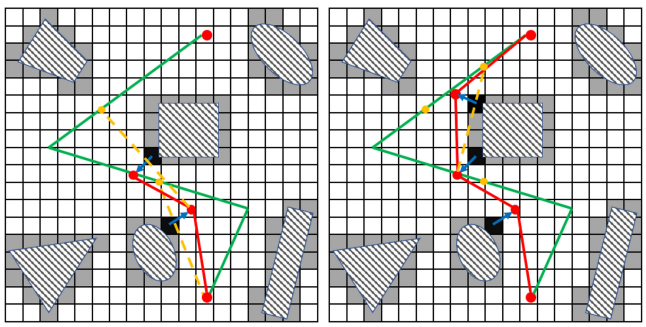


Fig. 9. A detoured and long path (green line) is shortened. The yellow discretized point is not visible to the last waypoint of the shortened path (red). The center points of the associated blocking voxel (black) are pushed away and appended as new waypoints.

in the direction orthogonal to l_d and coplanar to both l_d and the ESDF gradient at p_b (Line 6), after which it is appended to P_s (Line 7). The process continues until the last point is reached.

Algorithm 2: Finding a Topologically Equivalent Short-Cut Path P_s for P_r.

1 $P_d \leftarrow$ **Discretize**(P_r), $P_s \leftarrow \{P_d.\textbf{front}()\}$
2 **for each** $p_d \in P_d$ **do**
3 $l_d \leftarrow$ **Line**$(P_s.\textbf{back}(), p_d)$
4 **if** $\neg$ **LineVisib**(l_d) **then**
5 $p_b \leftarrow$ **BlockPoint**(l_d)
6 $p_o \leftarrow$ **PushAwayObs**(p_b, l_d)
7 $P_s.\textbf{push_back}(p_o)$
8 $P_s.\textbf{push_back}(P_d.\textbf{back}())$

Although in Algorithm 1, redundant connection between two guards is avoided, there may exist a small number of redundant paths between s and g [Fig. 8(d)]. To completely exclude repeated ones, we check the deformation between any two paths and only preserve topologically distinct ones. Also note that the number of distinctive paths grows exponentially with the number of obstacles. In case of complex environments, it is computationally intractable to use all paths to guide parallel optimization. For this reason, we only select the first $K_{\max}$ shortest paths. Paths more than $r_{\max}$ times longer than the shortest one are also pruned away. Such strategies bound the complexity and will not miss the potentially optimal solution, because a very long path is very unlikely to result in the optimal trajectory.

VI. Risk-Aware Trajectory Refinement

The trajectory refinement takes the best trajectory from the parallel PGO $\mathbf{p}_i(t)$ as input, modifies it in its vicinity, and outputs the refined trajectory $\mathbf{p}_r(t)$ (detailed in Algorithm 3). Central to it is the incorporation of the view and safety constraints into a trajectory optimization. The method starts by checking the visibility status (Section VI-A) of $\mathbf{p}_i(t)$, which retrieves the essential information required by the optimization. Given that safety cannot be guaranteed initially (Section VI-B1), the constraints are enforced through an iterative process to increase the visibility and reaction distance (Section VI-B2).

A. Checking Visibility Status

The visibility status is encoded by several variables: $t_f, \mathbf{p}_f$, $t_c, \mathbf{p}_c, \mathbf{v}_c$, which are important information about the unknown space passed through by $\mathbf{p}_i(t)$. Some of the involved variables are illustrated in Fig. 10.

1) Frontier Intersecting Point: As the unknown environment is only partially observed, at some time t_f, the trajectory $\mathbf{p}_i(t)$ exits the known-free space and enters the unknown space. The position $\mathbf{p}_f = \mathbf{p}_i(t_f)$, should be prioritized for observation due to three reasons: First, it is highly relevant for the future flight, because it belongs to a promising trajectory going toward the goal. Second, it may be dangerous to the flight. In the worst case an unknown obstacle can be right adjacent to it. Third, it will be reached earlier compared to other unknown position along $\mathbf{p}_i(t)$. Therefore, in **FrontierIntersection**(), we search along

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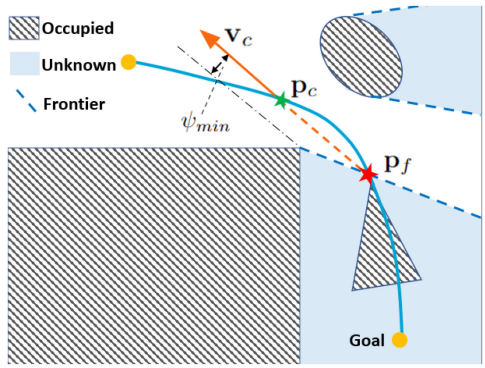


Fig. 10. An illustration of checking the visibility status of the inputted trajectory.

$\mathbf{p}_i(t)$ with a discrete time step, recording the first unknown point and the corresponding time.

2) Visibility Metric: During the flight, it is preferred that $\mathbf{p}_f$ becomes visible to some preceding positions on $\mathbf{p}_i(t)$. Quantitatively, we want some visibility level ψ of $\mathbf{p}_f$ to be not less than an expected level $\psi_{\min}$, so that not only $\mathbf{p}_f$ is visible but also the visibility level is tolerant to external disturbance. To gauge how reliable p_f is visible, we adopt the metric used by [35], [43], defining the visibility level from a position $\mathbf{p} \in \mathbb{R}^3$ to $\mathbf{p}_f$ as

$$\psi(\mathbf{p}, \mathbf{p}_f) = \min_{\mathbf{q} \in l(\mathbf{p}, \mathbf{p}_f)} d(\mathbf{q}) \tag{5}$$

which represents the the smallest Euclidean signed distance between the line segment $l(\mathbf{p}, \mathbf{p}_f)$ and obstacles. The evaluation of (5) requires traversing $l(\mathbf{p}, \mathbf{p}_f)$ and checking the Euclidean signed distance for each point. Fortunately, an ESDF derived from the occupancy grid map is maintained by our mapping module for supporting the trajectory optimization (Section IV), so minimal distance can be queried in constant time.

3) Critical View Direction: We are interested in whether $\mathbf{p}_f$ can be viewed from preceding positions on $\mathbf{p}_i(t)$, i.e., if $\psi(\mathbf{p}_i(t_k), \mathbf{p}_f) \geq \psi_{\min}$ for some time $t_k \leq t_f$. The best case is that for all $t \leq t_f$, the visibility level is higher than $\psi_{\min}$, in which no modification to the trajectory is needed (Lines 3 and4). However, in cluttered environments, $\mathbf{p}_f$ may not be observable until some time t_c, which means $\psi(\mathbf{p}_i(t), \mathbf{p}_f) < \psi_{\min}$ when $t < t_c$ and $\psi(\mathbf{p}_i(t), \mathbf{p}_f) \geq \psi_{\min}$ when $t \geq t_c$. Here, the view direction $\mathbf{v}_c = \frac{\mathbf{p}_c - \mathbf{p}_f}{\|\mathbf{p}_c - \mathbf{p}_f\|}$ where $\mathbf{p}_c = \mathbf{p}_i(t_c)$ is a critical view direction, in which $\psi(\mathbf{p}_i(t), \mathbf{p}_f)$ just reaches the desired level $\psi_{\min}$. In this case, **CriticalView**() reports $\mathbf{v}_c$ and the corresponding position $\mathbf{p}_c$ and time t_c.

B. Iterative Refinement

As is the worst case, an unknown obstacle may be revealed right behind $\mathbf{p}_f$ and block the trajectory. To ensure safety, we refine $\mathbf{p}_i(t)$ so that under the worst case, the quadrotor will be able to avoid collision by taking some maneuvers whose single-axis acceleration does not exceed the limit $a_{\max}$. We first check whether $\mathbf{p}_i(t)$ satisfies the worst-case safety criteria (Lines 5–7). If it does not, the critical view and safe reaction distance constraints are enforced (Lines 8–15).

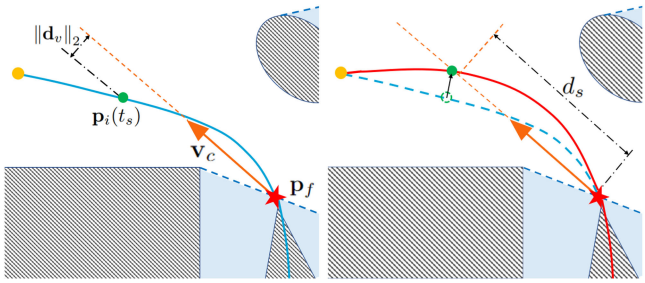


Fig. 11. The initial trajectory (blue) gets refined with the visibility status. Along the refined trajectory (red), $\mathbf{p}_f$ will become viewable earlier, wherein the reaction distance for collision avoidance is also increased.

Algorithm 3: Risk-aware Trajectory Refinement.

1 $\mathbf{p}_f, t_f \leftarrow$ **FrontierIntersection**$(\mathbf{p}_i(t))$
2 $visible, \mathbf{v}_c, \mathbf{p}_c, t_c \leftarrow$ **CriticalView**$(\mathbf{p}_i(t), \mathbf{p}_f)$
3 **if** $visible == True$ **then**
4 **return** $\mathbf{p}_i(t)$
5 $v_c \leftarrow \|\dot{\mathbf{p}}_i(t_c)\|_2$, $d_{cf} \leftarrow \|\mathbf{p}_c - \mathbf{p}_f\|_2$
6 **if** $v_c^2/2a_{max} \leq d_{cf} - R_q$ **then**
7 **return** $\mathbf{p}_i(t)$
8 $\hat{v}_s \leftarrow$ **AverageSpeed**$(\mathbf{p}_i(t), t_0, t_f)$
9 **repeat**
10 $\hat{d}_s \leftarrow \hat{v}_s^2/2a_{max} + R_q$
11 $t_s \leftarrow \arg\min_t f_{\mathbf{d}_v} + w_r f_{\mathbf{d}_{sf}}$
12 $\mathbf{p}_r(t) \leftarrow$ **RefineTrajectory**$(\mathbf{p}_i(t), \mathbf{p}_f, \mathbf{v}_c, t_s, \hat{d}_s)$
13 $v_s \leftarrow \|\dot{\mathbf{p}}_r(t_s)\|_2$, $d_{sf} \leftarrow \|\mathbf{p}_r(t_s) - \mathbf{p}_f\|_2$
14 $\hat{v}_s \leftarrow \alpha \cdot \hat{v}_s$
15 **until** $v_s^2/2a_{max} \leq d_{sf} - R_q$
16 **return** $\mathbf{p}_r(t)$

1) Worst-Case Safety Criterion: As shown in Section VI-A3, initially, $\mathbf{p}_f$ will not become reliably viewable until t_c at $\mathbf{p}_c$. Suppose at $\mathbf{p}_c$ the speed is v_c, while the distance to $\mathbf{p}_f$ is d_{cf}. Similar to [27], we check if (6) holds

$$v_c^2/2a_{\max} \leq d_{cf} - R_q \tag{6}$$

which means that if at $\mathbf{p}_c$ the quadrotor sees an obstacle, it can decelerate to a stop before colliding with the obstacle right behind $\mathbf{p}_f$. R_q compensates the quadrotor size and disturbance. If it is not true, extra constraints are added to meet this criterion.

2) View and Safety Constraints: If initially it would be too late for collision avoidance, $\mathbf{p}_f$ should be viewed at an earlier stage $t_s < t_c$, so that maneuvers to avoid collision can be taken in advance. Given the critical view direction $\mathbf{v}_c$, this constraint can be expressed as

$$\|\mathbf{d}_v\|_2 \leq \delta_v \tag{7}$$

$$\mathbf{d}_v = (\mathbf{p}_i(t_s) - \mathbf{p}_f) - ((\mathbf{p}_i(t_s) - \mathbf{p}_f) \cdot \mathbf{v}_c)\, \mathbf{v}_c. \tag{8}$$

The interpretation is that at t_s, the quadrotor should have already reached the ray $\{\mathbf{q} \,|\, \mathbf{q} = \mathbf{p}_f + \lambda \mathbf{v}_c, \lambda \geq 0\}$ from which $\mathbf{p}_f$ is viewable (Fig. 11). Besides the advanced observation, the distance from $\mathbf{p}_r(t_s)$ to $\mathbf{p}_f$ should be sufficient for an emergency

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stop

$$d_{sf} = \|\mathbf{p}_i(t_s) - \mathbf{p}_f\|_2 \geq d_s \tag{9}$$

where d_s is the safe reaction distance that depends on the speed v_s at the refined trajectory $\mathbf{p}_r(t_s)$: $d_s = v_s^2/2a_{\max} + R_q$. However, since the refined trajectory is not obtained yet, v_s is not available at this stage. To solve this chicken-and-egg problem, we introduce an iterative strategy. At the beginning we use the average speed of the segment of $\mathbf{p}_i(t)$ between $[t_0, t_f]$ as an estimate of v_s (Line 8), where t_0 is the start time of $\mathbf{p}_i(t)$. Then the trajectory is refined with the view and safety constraints (Lines 10–12). After the refinement, we check whether the new trajectory satisfies the safety criteria. If it does not, we increased the estimated speed $\hat{v}_s$ with a factor α slightly larger than 1 and redo the refinement (Lines 13–15). This strategy is complete because it only terminates after the safety criteria is indeed met. It also terminates quickly, since the speed along the smooth trajectory varies slightly and it only takes one or a few steps to find a good estimate of v_s in practice.

RefineTrajectory() is essentially optimizing the trajectory with the newly introduced view and safety constraints. To incorporate them into our gradient-based optimization (Section IV-B), (7)–(9) are soft-constrained by penalty functions

$$f_{\mathbf{d}_v} = \mathcal{F}(\delta_v, \|\mathbf{d}_v\|_2) \tag{10}$$

$$f_{\mathbf{d}_{sf}} = \mathcal{F}(\|\mathbf{p}_i(t_s) - \mathbf{p}_f\|_2, d_s) \tag{11}$$

Here $\mathcal{F}()$ is the penalty function (3). The Jacobian of these terms are

$$\frac{\partial f_{\mathbf{d}_v}}{\partial \mathbf{q}_k} = \begin{cases} \dfrac{2(\|\mathbf{d}_v\|_2 - \delta_v)\mathbf{d}_v^{\mathrm{T}}}{\|\mathbf{d}_v\|_2}(\mathbf{I} - \mathbf{v}_c\mathbf{v}_c^{\mathrm{T}})\dfrac{\partial \mathbf{p}_i(t_s)}{\partial \mathbf{q}_k}, & \|\mathbf{d}_v\|_2 > \delta_v \\ 0, & \text{else} \end{cases} \tag{12}$$

$$\frac{\partial f_{\mathbf{d}_{sf}}}{\partial \mathbf{q}_k} = \begin{cases} 0, & \|\mathbf{p}_i(t_s) - \mathbf{p}_f\|_2 \geq d_s \\ \dfrac{2(\|\mathbf{p}_i(t_s) - \mathbf{p}_f\|_2 - d_s)\mathbf{p}_i(t_s)^{\mathrm{T}}}{\|\mathbf{p}_i(t_s) - \mathbf{p}_f\|_2}\dfrac{\partial \mathbf{p}_i(t_s)}{\partial \mathbf{q}_k}, & \text{else} \end{cases}. \tag{13}$$

Here $\mathbf{q}_k$ B-spline control points associated with $\mathbf{p}_i(t_s)$. The refinement minimizes the cost function

$$f_r = \underbrace{\lambda_s f_s + \lambda_c f_c + \lambda_d (f_v + f_a)}_{f_{p2}} + \lambda_r (f_{\mathbf{d}_v} + w_r f_{\mathbf{d}_{sf}}) \tag{14}$$

where f_{p2} is exactly the same as (2). Note that applying the constraints to different t_s generates different results. However, including t_s into the optimization to find its best value would make the problem too difficult to be solved quickly. For simplicity, we first determine t_s as the time that minimizes $f_{\mathbf{d}_v} + w_r f_{\mathbf{d}_{sf}}$ for the initial $\mathbf{p}_i(t)$ and use it for the refinement (Lines 11 and 12). Quantitatively, the selected t_s leads to minimal violation of the constraints, therefore enforcing the constraints at t_s also requires less modification to $\mathbf{p}_i(t)$, which is a reasonably good choice.

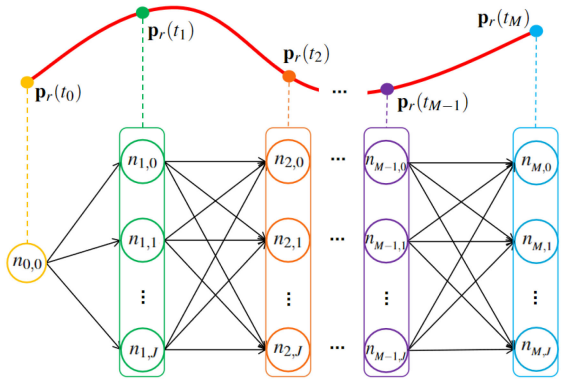


Fig. 12. The directed graph modeled for searching a sequence of yaw angles along the refined trajectory. Each layer of nodes (with the same color) is associated with a position on the trajectory.

VII. Yaw Angle Planning

Quadrotors typically have limited sensor FOVs. To improve flight safety, we plan the trajectory of yaw angle to actively observe the environments. Inspired by the recent two-step quadrotor motion planning paradigm [4]–[8], [15], we also decompose the yaw angle planning into a graph search problem and a trajectory optimization.

A. Graph Search

1) Problem Modeling: We model a graph search problem to seek for a sequence of yaw angles $\Xi := \{\xi_0, \xi_1, \ldots, \xi_M\}$ along the refined trajectory that trades off the smoothness and information gain (IG, detailed in Section VII-A2) of the unknown space. Given the quadrotor trajectory $\mathbf{p}_r(t)$ (Section VI), a set of positions $\mathbf{p}_{r,i}, i \in [0, 1, \ldots, M]$ uniformly distributed along the trajectory at $\{t_0, t_1, \ldots, t_M\}$ is selected. At $\mathbf{p}_i$ expect $i = 0$, where the yaw angle is already determined by the current quadrotor's state, several graph nodes $n_{i,j}, j \in [0, 1, \ldots, J]$ are created, each of which associates a different angle $\xi_{i,j}$ and the IG $g_{i,j}$ at the state $(\mathbf{p}_{r,i}, \xi_{i,j})$. For each pair of nodes n_{i,j_1}, n_{i+1,j_2} associated with adjacent positions, a graph edge from n_{i,j_1} to n_{i+1,j_2} is created. This process constructs a directed graph as shown in Fig. 12. To find a sequence of yaw that maximizes IG and smoothness, we search a path in the graph that minimizes the cost $c(\Xi)$, which can be solved efficiently by the Dijkstra algorithm

$$c(\Xi) = \sum_{i=1}^{M} \left\{ -g_{i,j_i} + \mu \left(\xi_{i,j_i} - \xi_{i-1,j_{i-1}}\right)^2 \right\} \tag{15}$$

where μ is used to adjust the weighting of smoothness.

2) Information Gain: We employ a similar method to [44], [45] which assesses potential IG as the number of unmapped voxels that comply with the camera model and are visible (not blocked by occupied voxels). However, the original method does raycasting for every voxel inside the camera FOV to validate their visibility, which is too expensive to function online. Therefore, we adapt it to better suit the real-time planning with several acceleration methods: *1) Memorizing visibility*: We borrow the techniques from [46] to avoid repeated raycasting. As depicted

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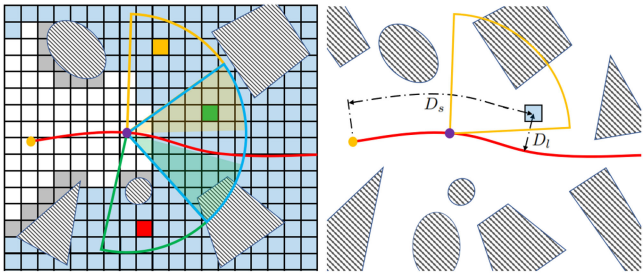


Fig. 13. (a) Evaluation of potential information gain. Blue transparent voxels represent unknown areas. Occupied and free space is represented by gray and white colors. The yellow voxel will contribute to the information gain, while the red voxel will not since it is blocked by occupied voxels. The green voxel is inside the overlapping region of multiple FOVs, whose visibility is stored after the first raycasting and queried directly in subsequent checks. (b) Weighting an unknown and visible voxel by its lateral and longitudinal distance to the trajectory.

in Fig. 13(a), we notice that at one position $\mathbf{p}_{r,i}$ where different $\xi_{i,j}$ are assessed, many voxels are in overlapping areas and are checked for visibility more than once. To avoid unnecessary repetition, we store the visibility of each voxel when it is checked for the first time, so that in subsequent check, the visibility can be queried directly. *2) Parallel evaluation*: The gains of different $\xi_{i,j}$ are computed in parallel. *3) Subsampling*: As is in [38], voxels inside the FOV are subsampled to approximate the actual gain, which leads to only a slight error but great run time reduction. In these ways, the overall IG evaluation time is reduced by over two orders of magnitude, as shown in Section VIII-C2.

In the context of exploration [44], every voxel contributes equally to the IG of one quadrotor configuration, ensuring that all space can be covered by the sensors uniformly.

Different from it, in point-to-point navigation, we do not aim at full coverage. In some cases, it is impossible to cover all unknown space around the trajectory within the dynamic capability, such as there are multiple unknown regions on opposite sides of the trajectory. More importantly, note that not all of the unknown regions are equally important to the flight. In particular, the ones closer to the trajectory and the current position of the quadrotor have higher influence, on which more emphasis should be placed. Therefore, we use (16) to bias the IG contribution Ig of a visible voxel centered at $\mathbf{m}$

$$Ig(\mathbf{m}) = \exp\left\{-w_l\, D_l(\mathbf{m}, \mathbf{p}_r(t)) - w_s\, D_s(\mathbf{m}, \mathbf{p}_r(t))\right\} \quad (16)$$

where $D_l(\mathbf{m}, \mathbf{p}_r(t))$ and $D_s(\mathbf{m}, \mathbf{p}_r(t))$ represent the lateral and longitudinal distance to trajectory $\mathbf{p}_r(t)$ [Fig. 13(b)].

B. Yaw Angle Optimization

Given the optimal path Ξ searched through the graph, we compute the trajectory of yaw angle $\phi(t)$ that is smooth, dynamically feasible and passes through the sequential angles ξ_j. We parameterize $\phi(t)$ as a uniform B-spline with control points $\Phi := \{\phi_{c,0}, \phi_{c,1}, \ldots, \phi_{c,N_c}\}$ and knot span δt_ϕ. In this way, the convex hull property can be employed to ensure dynamic feasibility [15]. The problem is formulated as

$$\begin{aligned}\arg\min_{\Phi} &\int_{t_0}^{t_M} (\dddot{\phi}(t))^2 dt + \gamma_1 \sum_{i=0}^{M} (\phi(t_i) - \xi_i)^2 \\ &+ \gamma_2 \left\{ \sum_{j=0}^{N_c-1} \mathcal{F}(\dot{\phi}_{\max}, |\dot{\phi}_{c,j}|) + \sum_{j=0}^{N_c-2} \mathcal{F}(\ddot{\phi}_{\max}, |\ddot{\phi}_{c,j}|) \right\}.\end{aligned} \quad (17)$$

Here the first term represents smoothness and the second term is a soft waypoint constraint enforcing $\phi(t)$ to pass through Ξ. The last two terms are the soft constraints for dynamic feasibility, wherein $\dot{\phi}_{c,j}$ and $\ddot{\phi}_{c,j}$ are the B-spline control points of angular velocity and acceleration

$$\dot{\phi}_{c,j} = \frac{\phi_{c,j+1} - \phi_{c,j}}{\delta t_\phi}, \quad \ddot{\phi}_{c,j} = \frac{\phi_{c,j+2} - 2\phi_{c,j+1} + \phi_{c,j}}{\delta t_\phi^2}. \quad (18)$$

Thanks to the convex hull property of B-spline, the entire trajectory is guaranteed to be feasible given that the control points do not exceed the dynamic limits $\dot{\phi}_{\max}, \ddot{\phi}_{\max}$.

VIII. Results

A. Implementation Details

We present tests in both real world and simulation. In real-world experiments, a customized quadrotor platform equipped with an Intel RealSense Depth Camera D435 is used. All the state estimation, mapping, planning, and control modules run on an Intel Core i7-8550 U CPU. For simulation, we use a simulating tool containing the quadrotor dynamics model, random map generator, and depth image renderer. The dynamics model relies on a numeric ODE solver *odeint*.[5] The depth images are rendered in GPU by projecting point cloud of the surrounding obstacles onto the image plane. Random noises are added to them to better mimic the real measurements. All simulations run on an Intel Core i7-8700 K CPU and GeForce GTX 1080 Ti GPU. The trajectory optimization is solved by a general nonlinear optimization solver NLopt.[6]

1) Global Planning: We use the approach [3] to compute global reference trajectories. Note that we focus on evaluating the local replanning system, therefore, naive global trajectories are given, such as straight-line trajectory connecting the start and goal positions.

2) Volumetric Mapping: In all tests, the quadrotor starts with no prior knowledge of environments. A volumetric mapping framework [47] fuses the depth images from the stereo camera into a occupancy grid map. An ESDF is derived from the occupancy grid map using an efficient distance transform algorithm [48] to support the gradient-based optimization (Section IV) and visibility evaluation (Section VI-A2). Trilinear interpolation is also used to reduce the distance error induced by the discrete grid map.

[5][Online]. Available: www.boost.org/doc/libs/1_73_0/libs/numeric/odeint/doc/html/index.html

[6]https://nlopt.readthedocs.io/en/latest/

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TABLE I
GUIDING OPTIMIZATION WITH TOPOLOGICAL PATHS VS. UTILIZING KINODYNAMIC SEARCH

Density (obs./m^2)	Method	Search time (ms)		Opt. time (ms)		Energy (m^2/s^5)	
		Avg	Std	Avg	Std	Avg	Std
0.2	Ours	3.15	1.46	3.94	0.55	**83.07**	74.04
	[40]+Opt.	92.90	76.20	3.81	0.12	129.10	175.09
	[15]+Opt.	1.32	1.19	3.62	0.36	142.93	115.71
0.3	Ours	3.67	1.53	3.99	0.42	**128.81**	178.18
	[40]+Opt.	114.68	103.28	3.78	0.29	136.40	295.04
	[15]+Opt.	1.69	2.34	3.59	0.30	183.06	159.70
0.4	Ours	3.96	1.52	4.10	0.31	**203.92**	324.86
	[40]+Opt.	186.24	140.83	3.79	0.38	233.44	445.82
	[15]+Opt.	2.69	2.92	3.58	0.33	288.74	131.37

3) State Estimation and Control: We localize the drone by a robust visual-inertial state estimator [49] in real-world tests. In simulation, ground truth odometry is generated by the quadrotor dynamics model. We use a geometric controller [50] to track both the position and yaw trajectory.

The following evaluation is divided into three parts. The first two parts evaluate the topological path searching and perception-aware planning strategy, respectively. The whole replanning framework is tested in the third part.

B. Trajectory Optimization With Topological Paths

We compare our trajectory optimization strategy, i.e., guiding optimization with topologically distinctive paths, against two approaches that search kinodynamic initial paths [15], [40] for optimization. The first approach utilizes a resolution-complete kinodynamic search [40]. It generates motion primitives by solving an optimal control problem, inducing a finite lattice discretization on the state-space. The state lattice is then explored by a graph-search algorithm, which finds optimal trajectories in the discretized space. The second approach employs [15], which is similar to [40], but adopts a pruning scheme to remove unpromising primitives and accelerates the search. In each comparison, the kinodynamic paths generated by [15], [40] are optimized with B-spline optimization (Section IV). Our method is initialized with a straight-line reference trajectory, which is guided by topological paths (Section V) and optimized in two phases (Section IV). Three random maps with low, medium, and high obstacle densities are utilized, in each of which we randomly generate 100 start and target positions.

As shown in Table I, our approach generates trajectories with the lowest costs in all three maps, with competitive efficiency. Initializing optimization by [15] is most efficient; however, its solutions are less promising. On the one hand, [15] searches with discretized control efforts, so the quality of solutions are restricted by the resolution of discretization. On the other hand, it prunes away numerous primitives during the search, which leads to the lack of optimality guarantee. Unlike it, [40] searches in the complete state lattice. This is more promising, but is also much more expensive. Meanwhile, [40] is limited by the discretization as well, making the initial path not necessarily lie near the global optimum. As indicated by the statistics, utilizing multiple distinctive paths is more advantageous for exploring the solution space thoroughly than the benchmarked methods, thanks to which trajectories with lower costs are obtained in all scenarios.

TABLE II
FOUR PLANNERS TESTED IN REAL-WORLD EXPERIMENTS

Planner	Strategy	Success Number	
		Scene 1	Scene 2
A	Optimistic & Velocity-tracking	0	0
B	Optimistic & Active exploration	0	0
C	Risk-aware & Velocity-tracking	2	0
D	Risk-aware & Active exploration	**3**	**3**

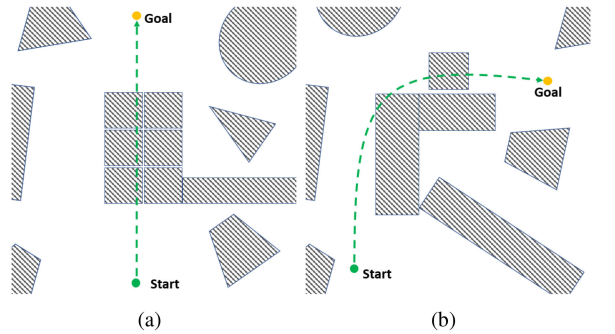


Fig. 14. An illustration of the two real-world scenarios for testing the perception-aware replanning. (a) A large obstacle is placed along the straight-line global reference trajectory. (b) An obstacle is placed right behind the corner.

C. Perception-Aware Planning Strategy

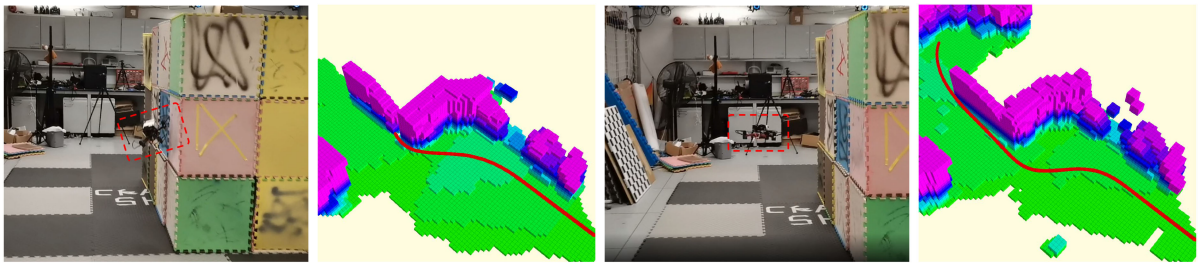
1) Real-World Tests: We conduct comparative experiments to show the importance of introducing active perception. Specifically we compare the proposed strategy: the risk-aware refinement (Section VI) and active exploration yaw (Section VII) with the commonly used ones: the optimistic assumption and the velocity-tracking yaw. Optimistic assumption treats all unknown space as collision-free, which is frequently adopted such as in [7], [14], [15], and [26]. The velocity-tracking yaw relates the desired yaw angle to the velocity: $\phi(t) = \arctan(\frac{v_y(t)}{v_x(t)})$, to increase the chance of seeing obstacles. Four local planners listed in Table II are tested in two scenes. Each planner is tested 3 times in both scenes and we record the number of successful flights. The maximum velocity and acceleration are set as 3 m/s and 2.5 m/s^2. Whenever collision along the trajectory is detected and the collision point is closer than 0.5 m, emergency stop[7] is conducted immediately for safety.

In the first scene, a straight-line global reference trajectory is given [Fig. 14(a)]. A large obstacle consisting of several boxes and boards are placed on the way. When the quadrotor approaches the obstacle, boxes in the front row will be revealed first, while others behind them are occluded and invisible at the beginning. Planners with optimistic assumption (A and B) are unaware of the potential danger behind. They simply replan trajectories to avoid the viewed boxes, along which there is low visibility to the boxes in the back, as shown in Fig. 16(a). As a result, the quadrotor gets "surprised" by the occluded boxes afterward and pauses in emergency, as shown in Fig. 15(a) and

[7]The quadrotor fully utilizes its thrust to decelerate to a stop as fast as possible.

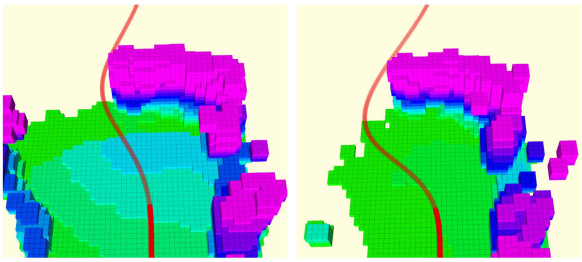
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(a) (b) (c) (d)

Fig. 15. Comparison of four local planners in scene 1. (a) Using planners A and B, the boxes in the back row are not discovered until the quadrotor gets very close. Consequently, the quadrotor has to pause in emergency. (b) The 3D maps (colorful voxels) and executed trajectories (red curves) at the moment when boxes in the back are just discovered. (c) and (d) Quadrotor avoid all boxes successfully with planners D. More details are shown in the video.



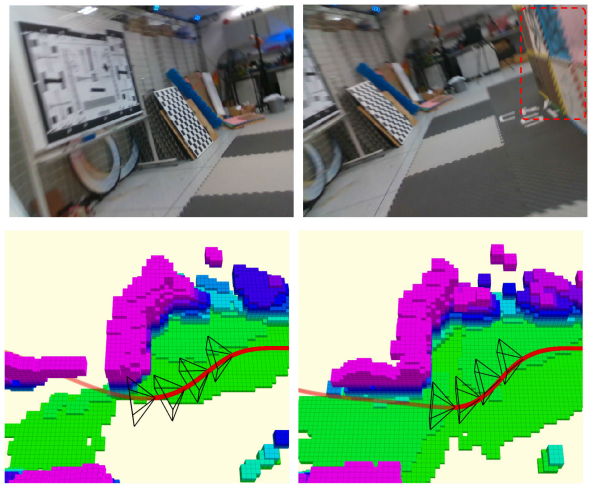
(a) (b)

Fig. 16. A comparison of the trajectories in scene 1 replanned with (a) optimistic assumption and (b) risk-aware refinement. Trajectories already executed and not executed yet are shown in opaque red and transparent red, respectively. (a) Along the trajectory, visibility toward the unknown region behind the observed obstacle is poor. (b) The trajectory deviates more laterally; therefore, along it there is greater visibility toward the unknown area in the back.

(b). In contrast, planners with risk awareness (C and D) generate trajectories that deviate a bit more laterally, along which visibility toward the unknown area in the back is higher [Fig. 16(b)]. Therefore, in both cases, the quadrotor reaches the goal more times. However, with the velocity-tracking yaw (planner C), the quadrotor does not face toward the unknown area in the back quickly, which postpones the discovery of occluded boxes and causes 1 failure. In comparison, with the active exploration yaw (planner D), the quadrotor quickly turns toward the unknown area and observes the previously occluded boxes, enabling itself to take action earlier. This comparison is displayed in Fig. 17.

In the second scene, an obstacle is placed right behind the corner, which is invisible to the quadrotor until it turns right. The reference trajectory is set to pass through the obstacles deliberately [Fig. 14(b)]. In this scene, the quadrotor can only reach the goal safely with planner D. For the other three planners, the quadrotor collides with the obstacle behind the corner, due to either the poor visibility of the replanned trajectories (planners A and B), or the delay of perception caused by the velocity-tracking yaw (planner C). Note that even emergency stop is conducted, the quadrotor fails to avoid collision in time (Fig. 18). The comparisons of the replanned trajectories and yaw angle are displayed in Figs. 19 and 20(b), respectively.

The experiments demonstrate two critical factors to survive in high-speed flights: (a) having good visibility toward the unknown regions that will influence the flight and (b) looking toward the relevant direction to eliminate those unknown regions



(a) (b)

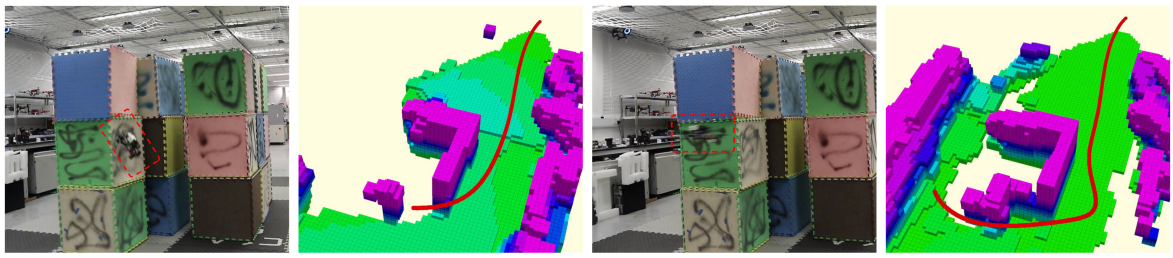
Fig. 17. A comparison in scene 1 of (a) velocity-tracking yaw and (b) active exploration yaw. Top: The first person view (FPV) images. Bottom: the 3D maps and trajectories associated with the FPV images. The camera poses within 0.9 s (interval = 0.3 s) are also visualized. (a) The quadrotor face to the left side at the beginning, so unknown boxes on the right side are not mapped completely. (b) The quadrotor discovers the previously occluded boxes earlier.

actively. The proposed method takes into account these factors and guarantees safety for fast flight. More details about the experiments are presented in the attached video.

2) Analysis: We quantitatively study the effect of introducing perception-aware planning. Specifically, flight tests are performed in various obstacle densities, with the perception-aware planning disabled/enabled. $v_{\max} = 3$ m/s and $a_{\max} = 2.5$ m/s^2 are set for all flights. During the flights, whenever a collision along the trajectory in execution is detected, the distance between the centroid of the quadrotor and the closest collision point (distance to collision, D2C) is recorded. Here we use D2C as the measure of how much the quadrotor is "surprised" when discovering collisions, i.e., a lower D2C means a smaller safety margin and a higher level of "surprise." The box plot of D2Cs is displayed in Fig. 21. From the statistics, it is clear that the perception-aware planning significantly reduce the level of "surprise." Without the proposed method, the minimal D2Cs quickly drop to near zero as the density increases, putting the quadrotor in great hazard.

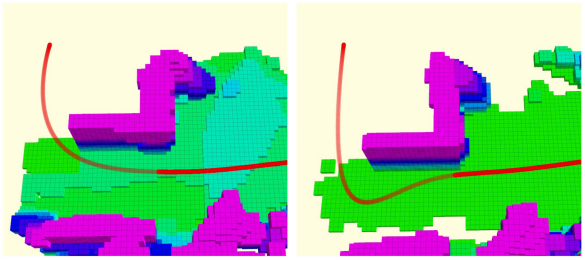
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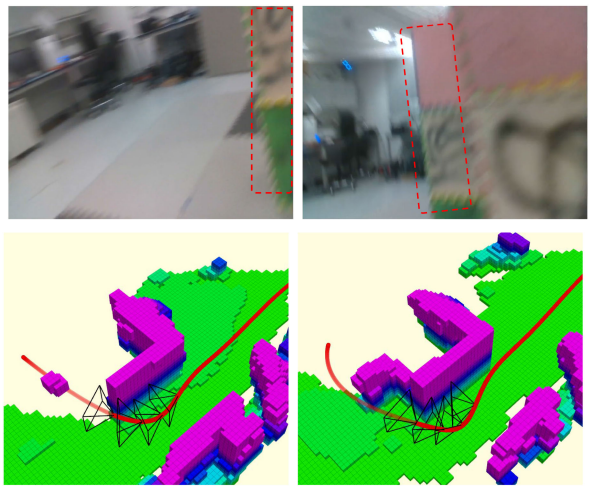
(a) (b) (c) (d)

Fig. 18. Tests in scene 2. (a) and (b) Quadrotor fails to react to the obstacle after turning right and crashes. (c) and (d) Quadrotor avoids the obstacle behind the corner successfully.



(a) (b)

Fig. 19. A comparison of the trajectories in scene 2 with (a) optimistic assumption and (b) risk-aware refinement. (a) The trajectory is close to the corner. (b) The trajectory is slightly longer around the corner, where the quadrotor can observe the unknown space in the back and react to possible dangers.



(a) (b)

Fig. 20. (b) A comparison in scene 2 of (a) velocity-tracking yaw and (b) active exploration yaw. (a) The quadrotor turns its head to the right later. The hidden obstacles is not observed until getting close. (b) The quadrotor looks to the right side earlier and discovers the obstacle behind the corner.

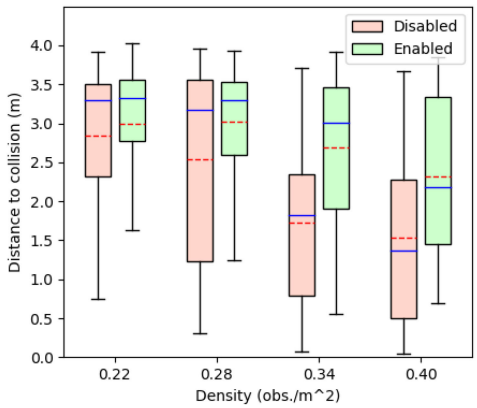
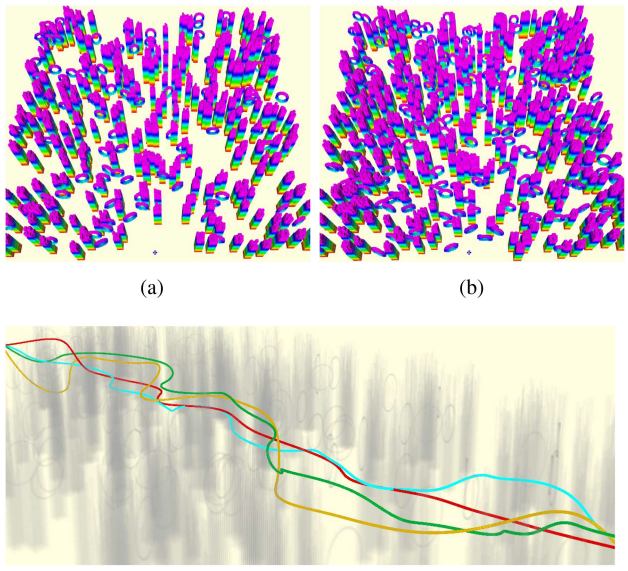


Fig. 21. Distance to collision with/without perception-aware planning. The median and mean values are shown by blue solid and red dash lines.



(a) (b)

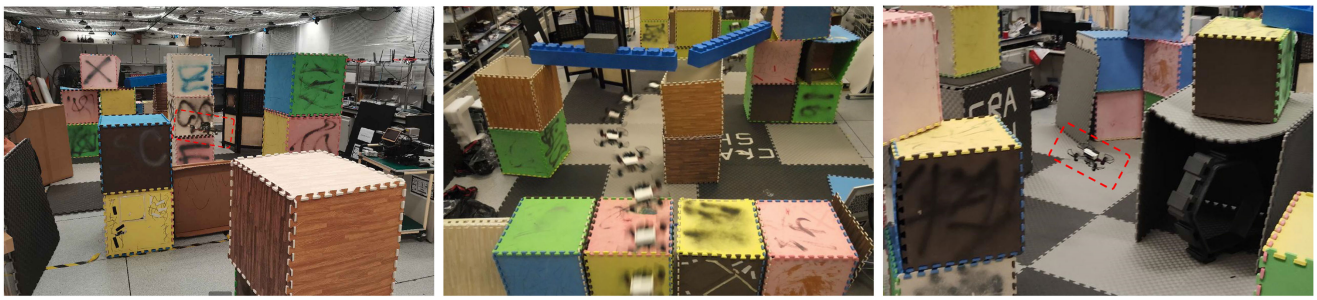
(c)

Fig. 22. Samples of the maps and generated trajectories. (a) Sample map 1: 0.25 obs./m ^{2}. (b) Sample map 2: 0.4 obs./ m^2. (c) Trajectories generated by the proposed method (red), FASTER [10] (green), EWOK [14] (cyan), and RE Traj. [15] (yellow). Obstacles are set as gray transparent for clarity.

As mentioned in Section VII-A2, the evaluation of information gains (IGs) [44], [45] can take up enormous computational time. We examine our methods adopted to accelerate their computation, including memorizing visibility (M), parallel evaluation (P), and subsampling (S) (Section VII-A2). Table III records the time taken for IGs in every yaw angle planning. It can be seen that computing IGs can consume hundreds of milliseconds originally, which is prohibited for fast navigation. Fortunately, with our adaptations the computation time of IGs is reduced by over two orders of magnitude, making IG applicable for real-time replanning. Although the subsampling method may yield a slight estimation error of IGs, the error was shown to be acceptable [38].

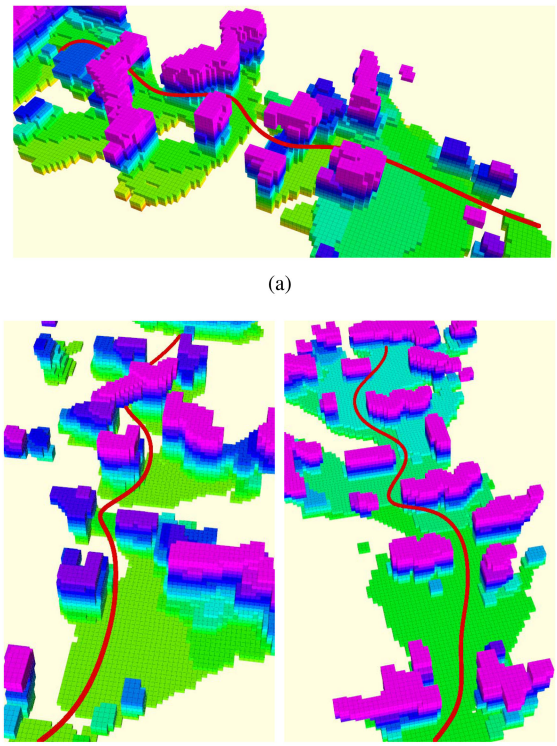
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(a) (b) (c)

Fig. 23. Three indoor scenes for the fast flight experiments. Several tests are conducted for each scene; more details are available in the attached video. (a) The quadrotor passes a horizontal cardboard, after which it will avoid the vertical pillar. (b) Composite image of the flight experiment. (c) The quadrotor flies in the narrow passages and avoids boxes.



(a) (b) (c)

Fig. 24. Samples of the online generated maps and trajectories executed by the quadrotor in the different indoor scenes.

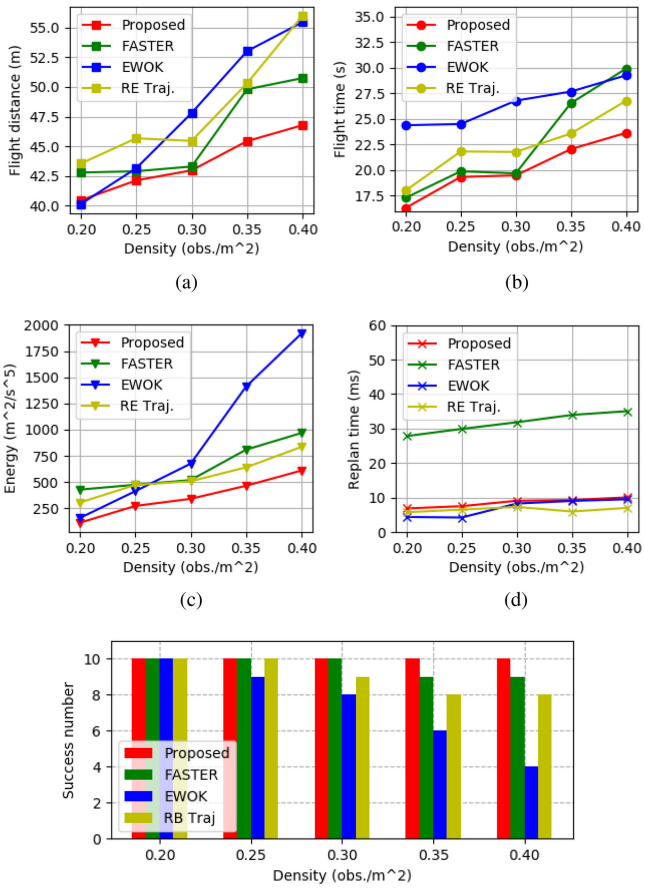


Fig. 25. Results of the comparisons between the proposed method and FASTER, EWOK, and RE Traj. (a) Avg. flight distance (m). (b) Avg. flight time (s). (c) Avg. consumed energy (m^2/s^5). (d) Avg. replan time (ms). (e) Number of successful flights.

3) Benchmark Comparisons: We compare the proposed strategy with the safe local exploration (SLE) presented in [38] in simulation. This strategy originated from the "next-best-view" planner [44] in the exploration literature, but is adapted for online functioning in goal-reaching tasks. It repeatedly selects intermediate goals that are closer to the final goal and have higher information gain, after which a local planner replans new trajectories toward the goals. It also adopts the velocity-tracking yaw, as is detailed in Section VIII-C1. To compare the strategies fairly, we integrated both of them with our robust optimistic replanning (Sections IV and V). They are tested in 10 random maps with 5 different obstacle densities, 5 trials are conducted for each map. We compare the number of successful flights, flight time, and flight distance. Samples of the maps are displayed in Fig. 22(a) and (b).

As is shown in Table IV, the proposed strategy achieves a higher number of successful flights when the scene gets more cluttered. Our strategy enforces visibility to dangerous unknown areas, and control the yaw angle to observe those areas actively. Therefore, it can guarantee safety even the environment becomes

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(a) (b) (c)

Fig. 26. Autonomous fast flight experiments in two dense forests. (a) Flying through forest 1. (b) Flying up the slope in forest 1. (c) Flying through forest 2.

TABLE III
COMPUTATION TIME OF INFORMATION GAINS WITH DIFFERENT ACCELERATION METHODS

Acceleration Method		None	M	M+P	M+P+S
Computation time (ms)	Avg.	406.213	195.242	69.340	1.621
	Std.	165.284	71.749	22.415	0.499
	Max.	816.205	375.071	137.842	2.930

TABLE IV
COMPARISONS OF THE PERCEPTION-AWARE PLANNING STRATEGIES

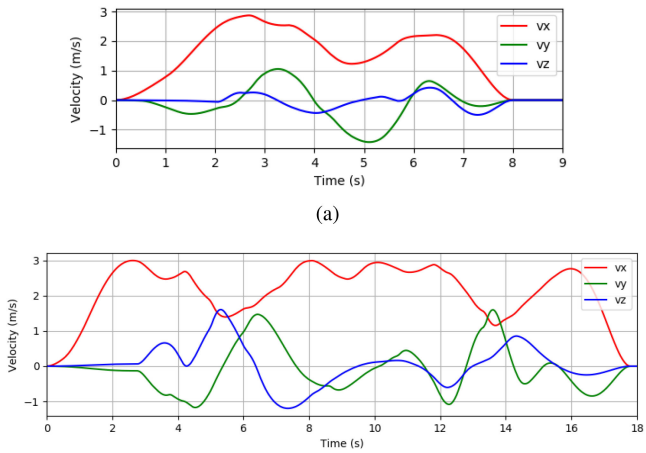
Density (obs./m^2)	Method	Number of Success	Flight Distance (m)	Flight Time (s)
0.2	SLE [38]	10	43.475	20.012
	Proposed	10	**41.822**	**16.608**
0.25	SLE [38]	10	44.682	20.593
	Proposed	10	**42.133**	**19.338**
0.3	SLE [38]	9	44.874	22.316
	Proposed	**10**	**42.982**	**19.483**
0.35	SLE [38]	9	46.093	23.577
	Proposed	**10**	**45.436**	**22.050**
0.4	SLE [38]	8	47.817	27.903
	Proposed	**10**	**46.776**	**23.639**

very complex. For SLE, the velocity-tracking yaw is the major cause of failure, as it may not face toward dangerous unknown regions in time, as has been shown in Section VIII-C1.

Besides, our strategy is also more beneficial to achieve lower flight distance and time than SLE. SLE only selects intermediate goals and plans within the known unoccupied space, which is conservative. Besides, since the selection of intermediate goals takes information gain into account, the quadrotor tends to take some detours to gather more information, which leads to longer flight distance and time. In contrast, our strategy plans in both the known and unknown space, allowing more aggressive behaviors under the premise of safety. Moreover, instead of treating all unknown areas equally, it only focuses on observing areas that are more relevant to the flight, which eliminates many unnecessary detours and improves the overall flight efficiency.

D. Replanning Framework for Fast Flight

1) Benchmark Comparisons: We compare our replanning framework with several state-of-the-art methods, FASTER [10], EWOK [14], and RE Traj. [15]. FASTER belongs to the hard-constrained category (Section II-A), and features maintaining a feasible and safe back-up trajectory in the free-known space at each replanning step to improve safety. It also adopts an MIQP formulation to obtain a more reasonable time allocation of the trajectories. Both [14], [15] belong to the gradient-based methods. They utilize a uniform B-spline trajectory representation to replan efficiently. Zhou *et al.* [15] further exploit the convex hull property of B-spline and introduce a kinodynamic path searching to find more promising initial trajectories. We also test the four methods in 10 random maps with five obstacle densities. Note that all benchmarked methods are open-source and we use their default parameter settings. The number of successful flights, average flight distance, flight time, energy (integral of squared jerk), computation time of each replanning, and total replan number in each flight are recorded. Samples of the maps and the trajectories generated by the four methods are shown in Fig. 22.



(a)

(b)

Fig. 27. Velocity profiles of sample flights in (a) indoor experiment (b) forest experiment.

As displayed in Fig. 25, our method outperform others in the aspects of flight distance, flight time, and energy consumption, with competitive computation efficiency. FASTER rarely fails in the tests, thanks to the back-up trajectories. However, due to the computationally demanding MIQP formulation, its overhead is higher. The other two benchmarked methods are more efficient.

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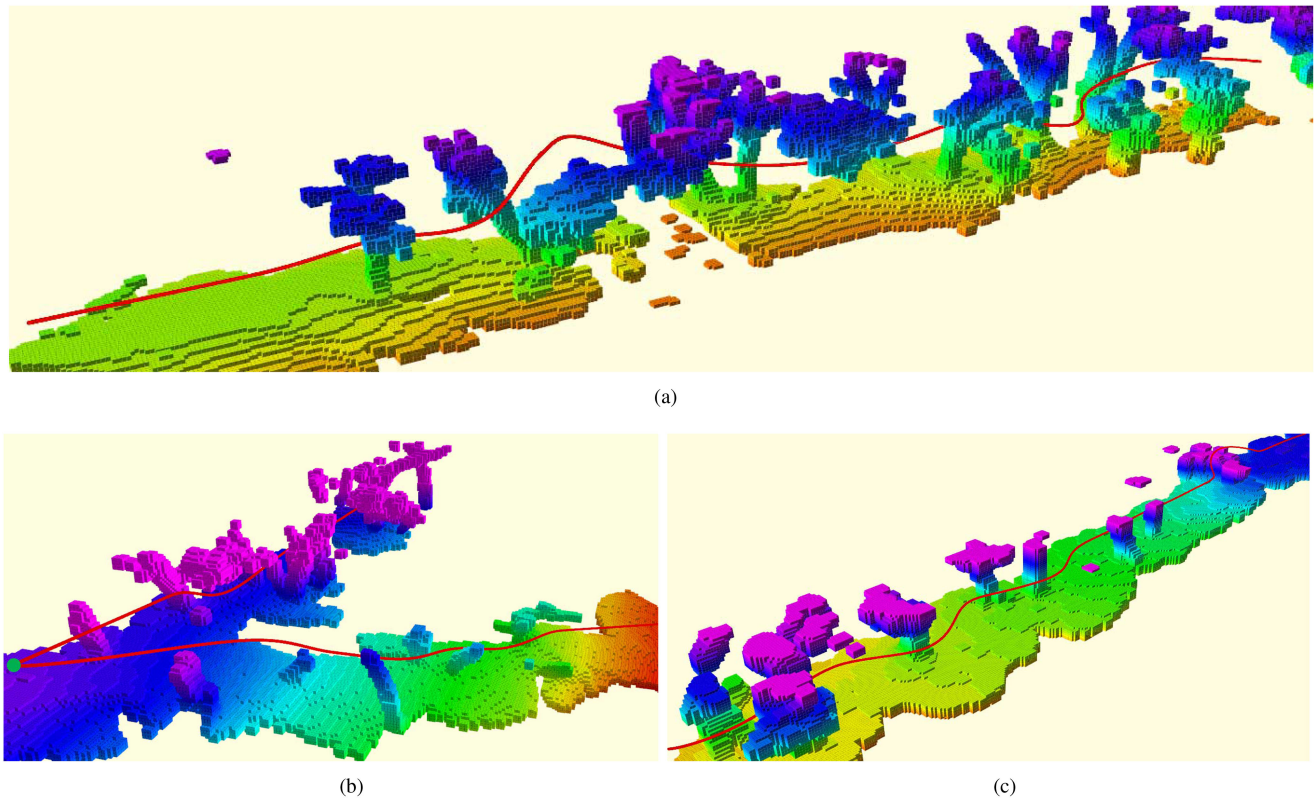


Fig. 28. Online generated maps of the forests and trajectories executed by the quadrotor. The map is colored by height. (a) Flight in forest 1. (b) The quadrotor flies up the slope to the first goal (green circle), after which it flies toward the second goal. (c) Flight in forest 2.

However, EWOK suffers from the local minima issue, so it usually fails or outputs low-quality solutions in dense environments. The kinodynamic path searching and B-spline optimization adopted by RE Traj. relieve the local minima significantly. Nonetheless, due to the lack of perception consideration, the succuss number is mediocre in dense environments. Compared to them, the proposed method searches the solution space effectively with the guidance of topologically distinctive paths, and generates high-quality trajectories consistently. Safety is also reenforced by introducing perception awareness.

2) Indoor Flight Test: We conduct aggressive flight experiments in three indoor scenes (Figs. 2 and 23) to validate our planning system. Various types of obstacles are placed randomly and densely to make up the challenging flight environments. Distances of neighboring obstacles are only around 1 m, making the space for safe navigation very limited. Besides, the high obstacle density makes visibility to the environment very restricted, since many obstacles are occluded by others, which poses greater challenges to the replanning algorithm.

In each experiment, the final goal is set to 14 m away from the quadrotor. Straight-line global reference trajectories are given and local replanning is conducted within a horizon of 7 m. Samples of the online generated map and executed trajectories are presented in Fig. 24. The velocity profile of one flight is shown in Fig. 27(a), in which the maximum speed is 2.90 m/s and average speed is 1.77 m/s. The flight distance and time are 14.12 m and 8.0 s, respectively. We refer the readers to the attached video for more tests.

3) Outdoor Flight Test: Finally, we conduct fast flight tests in three different outdoor scenes, as displayed in Fig. 26, to validate our planning method in natural environments. The outdoor environments are typically unstructured and irregular, where the quadrotor should perform agile 3D maneuvers to avoid obstacles such as rocks and branches and leaves of trees. Note that despite the outdoor environments, we do not use external devices for localization.

Results of the online generated map and executed trajectories are presented in Fig. 28. In the first scene, the quadrotor flies through the forest to the goal 39 m away from the initial position. The velocity profile is shown in Fig. 27(b). The maximum speed is 3.19 m/s and average speed is 2.29 m/s. The flight takes 40.78 m and 17.83 s. In the second scene, the quadrotor flies up a slope to the first goal, after which it flies to the second goal. The first goal is 30 m away and the change in height is 7 m. The second goal is 17 m far. The whole flight takes 48.71 m and 23.19 s. The third scene is a larger forest, where the goal is set to 45 m away. The flight distance is 46.85 m, which takes 21.91 s to finish. The maximum and average speed are 3.41 and 2.14 m/s. More details of the flights are shown in the attached video.

IX. Conclusion

In this article, we proposed a robust and perception-aware replanning method for high-speed quadrotor autonomous navigation. The PGO and topological path searching were devised to escape from local minima and explore the solution space more

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thoroughly, through which higher robustness and optimality guarantee are obtained. The robust planner was further enhanced by the perception-aware strategy, which takes special caution about regions that may be dangerous to the quadrotor. The yaw angle of the quadrotor was also planned to actively explore the environments, especially areas that were relevant to the future flight. The planning system was evaluated comprehensively through benchmark comparisons. We integrated the planning method with global planning, state estimation, mapping, and control into a quadrotor platform and conducted extensive challenging indoor and outdoor flight tests. Results showed that the proposed method was robust and capable of supporting fast and safe flights. We released the implementation of our system to the community.

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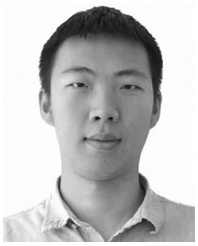
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Boyu Zhou received the B.Eng. degree in mechanical engineering from Shanghai Jiao Tong University, Shanghai, China, in 2018. He is currently working toward the Ph.D. degree in electronic and computer engineering at the Hong Kong University of Science and Technology, Hong Kong.

His research interests include aerial robots, autonomous navigation, motion planning, dense mapping, exploration, and swarm.



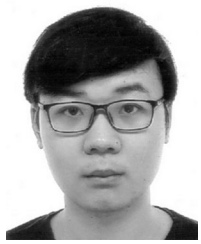
Jie Pan received the B.Eng. degree in software engineering from Xidian University, Xi'an, China, in 2017, and the M.Phil. degree in electronic and computer engineering from the Hong Kong University of Science and Technology, Hong Kong, in 2019.

He is currently a Research Assistant with the Department of Electronic and Computer Engineering, Hong Kong University of Science and Technology. His research interests include state estimation, sensor fusion, localization and mapping, and autonomous navigation in complex environments.



Shaojie Shen received the B.Eng. degree in electronic engineering from the Hong Kong University of Science and Technology, Hong Kong, in 2009, and the M.S. degree in robotics and the Ph.D. degree in electrical and systems engineering from the University of Pennsylvania, Philadelphia, PA, USA, in 2011 and 2014, respectively.

In September 2014, he joined the Department of Electronic and Computer Engineering, Hong Kong University of Science and Technology, as an Assistant Professor, and is promoted to Associate Professor in July 2020. His research interests include robotics and unmanned aerial vehicles, with a focus on state estimation, sensor fusion, computer vision, localization and mapping, and autonomous navigation in complex environments.



Fei Gao received the B.Eng. degree in control science and engineering from Zhejiang University, Hangzhou, China, in 2015, and the Ph.D. degree in electronic and computer engineering from the Hong Kong University of Science and Technology, Hong Kong, in 2019.

He is currently an Assistant Professor with the Department of Control Science and Engineering, Zhejiang University, where he co-directs the Field Autonomous System and Computing Laboratory. His research interests include unmanned aerial vehicles, autonomous navigation, motion planning, optimization, and localization and mapping.

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